Conditional Probability and Bayes' Theorem

1 Conditional Probability

1.1 Definition

Definition 1.1 (Conditional Probability)**.**

The conditional probability of an event A, given that the event B has occurred, is defined by

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

provided that $\mathbb{P}(B) \neq 0$.

Remark. It can be easily verified that the *conditional probability* defined above satisfies the *probability axioms*. Thus it is actually a *probability function* whose *sample space* is B.

The essence of conditional probability is that it is a corresponding probability function of a given probability function.

1.2 Multiplication Rule

Definition 1.2 (Multiplication Rule)**.**

The probability that two events, A and B, both occur is given by the multiplication rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) \quad \text{provided} \quad \mathbb{P}(A) > 0$$

or

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B) \quad \text{provided} \quad \mathbb{P}(B) > 0$$

2 Independent Events

2.1 Definition

Definition 2.1 (Independent Events)**.**

Events A and B are **independent** iff

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Otherwise, they are called **dependent** events.

2.2 Consequence

I When $\mathbb{P}(A) > 0$, we have

$$\mathbb{P}(B \mid A) = \mathbb{P}(B) \qquad \Longleftrightarrow \qquad A \text{ and } B \text{ are independent}$$

II When $\mathbb{P}(B) > 0$, we have

$$\mathbb{P}(A \mid B) = \mathbb{P}(A) \qquad \Longleftrightarrow \qquad A \text{ and } B \text{ are independent}$$

III A and B are independent, iff the following pairs of events are also independent:

- A and B'
- A' and B
- A' and B'

2.3 Mutually Independent Events

Definition 2.2 (Three Mutually Independent Events)**.**
Events A, B, and C are **(mutually) independent** iff the following two conditions hold:

- A, B, and C are **pairwise independent**, i.e., A and B, B and C, and A and C are independent, respectively.
- $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$

Definition 2.3 (Mutually Independence for More Events)**.**
$A_1, A_2, \cdots, A_n$ are **(mutually) independent** iff

$$\mathbb{P}\left(\bigcap_{i=1}^{k} A_i\right) = \prod_{i=1}^{k} \mathbb{P}(A_i)$$

holds for all $k = 2, 3, \cdots, n$.

3 Bayes' Theorem

Theorem 3.1 (Bayes' Theorem)**.**

Assume that

1. $B_1, B_2, \cdots, B_n$ *are* ***mutually exclusive*** *and* ***exhaustively*** *events.*
2. *The* ***prior probability*** *of all* B_i *is nonzero, i.e.,*

$$\mathbb{P}(B_i) > 0, \quad i = 1, 2, \cdots, n$$

Then, for any event A*, we have*

- ***Law of Total Probability****:*
$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$$
- ***Bayes' Rule****: If* $\mathbb{P}(A) > 0$*, then we have the* ***posterior probability*** *of* B_k*:*
$$\mathbb{P}(B_k \mid A) = \frac{\mathbb{P}(B_k \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \mid B_k)\mathbb{P}(B_k)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$$